6.4 TIME RESPONSE OF A SECOND ORDER CONTROL SYSTEM

A second order control system is one wherein the highest power of s in the denominator of its transfer function equals 2.

A general expression for the transfer function of a second order control system is given by

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad ...(6.14)$$

The parameters ζ and ω_n will be explained later. The block diagram representation of the transfer function given by Eq. (6.14) is shown in Fig. 6.4.1.

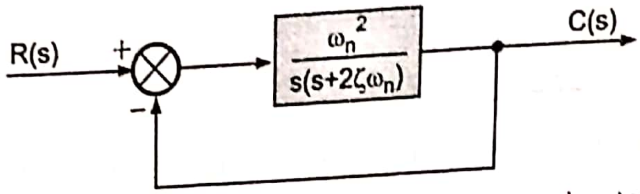

Fig. 6.4.1. Block diagram of a second order control system.

6.4.1. Time response of a second order control system subjected to unit step input function

The output for the system is given by

$$C(s) = R(s)\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad ...(6.15)$$

As the input is a unit step function

$$r(t) = 1 \quad \text{and} \quad R(s) = \frac{1}{s}$$

Therefore, substituting in Eq. (6.15)

$$C(s) = \frac{1}{s}\cdot\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad ...(6.16)$$

In Eq. (6.16) rewriting the term $(s^2 + 2\zeta\omega_n s + \omega_n^2)$ as $[(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)]$ and breaking R.H.S. into partial fractions

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{[(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)]} \qquad ...(6.17)$$

put $\quad \omega_d = \omega_n\sqrt{1 - \zeta^2}$

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2}. \qquad ...(6.18)$$

Eq. (6.18) can be rewritten as follows

$$C(s) = \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2} - \frac{\zeta\omega_n}{\omega_d}\cdot\frac{\omega_d}{(s + \zeta\omega_n)^2 + \omega_d^2} \qquad ...(6.18\ a)$$

Taking inverse Laplace transform on both sides of Eq. (6.18 *a*)

$$\mathcal{L}^{-1}C(s) = \mathcal{L}^{-1}\left[\frac{1}{s} - \frac{s+\zeta\omega_n}{(s+\zeta\omega_n)^2+\omega_d^2} - \frac{\zeta\omega_n}{\omega_d}\cdot\frac{\omega_d}{(s+\zeta\omega_n)^2+\omega_d^2}\right] \quad \text{...(6.19)}$$

$$\therefore \quad c(t) = 1 - e^{-\zeta\omega_n t}\,.\cos\omega_d t - \frac{\zeta\omega_n}{\omega_d}\,.\,e^{-\zeta\omega_n t}\,.\sin\omega_d t \quad \text{...(6.20)}$$

Since $\omega_d = \omega_n\sqrt{1-\zeta^2}$ Eq. (6.20) is rewritten as below

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left[(\sqrt{1-\zeta^2})\cos\omega_d t + \zeta\sin\omega_d t\right] \quad \text{...(6.21)}$$

put

$$\sin\phi = \sqrt{1-\zeta^2}, \quad \therefore \quad \cos\phi = \zeta$$

$$\therefore \quad c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}(\sin\phi\cos\omega_d t + \cos\phi\sin\omega_d t)$$

or

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi) \quad \text{...(6.22)}$$

Since $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $\phi = \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$

Eq. (6.22) is rewritten as below

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left[(\omega_n\sqrt{1-\zeta^2}t) + \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right] \quad \text{...(6.23)}$$

The error is given as $e(t) = r(t) - c(t)$

and

$$r(t) = 1$$

$$\therefore \quad e(t) = 1 - \left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_n\sqrt{1-\zeta^2}\,t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right]$$

or

$$e(t) = \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\,.\sin\left[\omega_n\sqrt{1-\zeta^2}\,t + \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right] \quad \text{...(6.24)}$$

The steady state error $e_{ss} = \lim\limits_{t\to\infty}\dfrac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\,.\sin\left[\omega_n\sqrt{1-\zeta^2}\,t + \tan^{-1}\left(\dfrac{\sqrt{1-\zeta^2}}{\zeta}\right)\right]$

The time response expression given by Eq. (6.23) indicates that for values of $\zeta < 1$ the
...onentially decaying oscillations having a frequency $\omega_n\sqrt{(1-\zeta^2)}$ and

ratio and the term $\zeta\omega_n$ is called damping factor or actual damping or damping coefficient.

The time response of a second order control system given by Eq. (6.23) is influenced by its damping ratio (ζ). The cases for the values of damping ratio as (*a*) $\zeta < 1$ (*b*) $\zeta = 0$ (*c*) $\zeta = 1$ (*d*) $\zeta > 1$ are considered below :

(*a*) The time response and the error in relation to Eqs. (6.23) and (6.24) for $\zeta < 1$ are plotted in Fig. 6.4.2 (*a*) and (*b*).

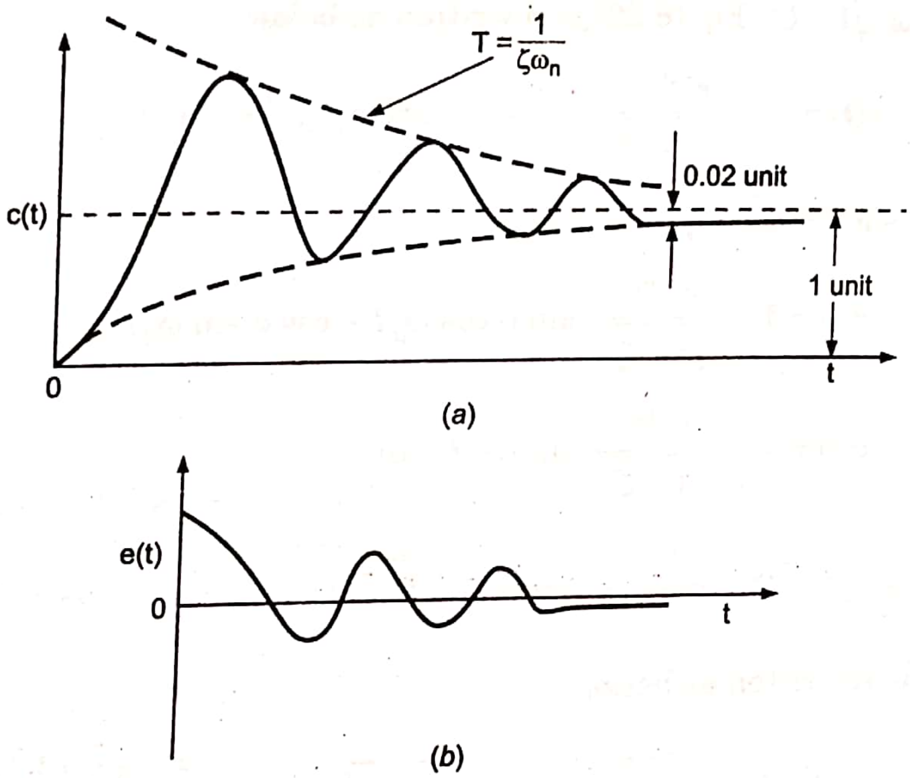

Fig. 6.4.2. Time response and error of a second order control system ($\zeta < 1$, under damped case) subjected to unit step input function.

As stated above, if $\zeta < 1$ the time response presents damped oscillations and such a response is called underdamped response.

The response settles within 2% of the desired value (1 unit) after damping out the oscillations in a time $4T$, where $T = 1/\zeta\omega_n$.

(*b*) An expression for the time response of a second order system having $\zeta = 0$ when subjected to unit step input function is derived hereunder :

Substituting $\zeta = 0$ in Eq. (6.23) following equation is obtained :

$$c(t) = 1 - \frac{e^{-o\omega_n t}}{\sqrt{1-o^2}} \sin\left[\omega_n \sqrt{1-o^2}\, t + \tan^{-1}\left(\frac{\sqrt{1-o^2}}{o}\right)\right]$$

or $$c(t) = 1 - \sin(\omega_n t + \tan^{-1}\infty)$$

or $$c(t) = 1 - \sin\left(\omega_n t + \frac{\pi}{2}\right)$$

or $$c(t) = (1 - \cos\omega_n t) \qquad \text{...(6.25)}$$

The time response in relation to Eq. (6.25) is plotted in Fig. 6.4.3 which indicates sustained (undamped) oscillations.

(*c*) An expression for the time response of a second order control system having $\zeta = 1$ when subjected to unit step input function is derived hereunder :

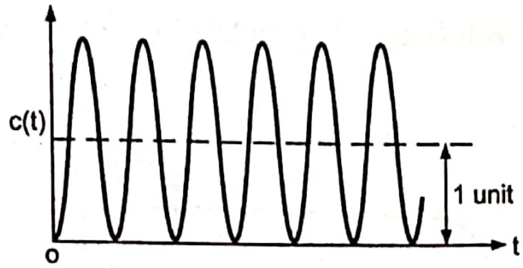

Fig. 6.4.3. Time response of a second order control system (ζ = 0, undamped case) subjected to unit step input functions.

Taking limits of Eq. (6.23) as the value of ζ approaches 1, the equation derived below is obtained

$$c(t) = \lim_{\zeta \to 1}\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_n\sqrt{1-\zeta^2}\,t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right]$$

In the above equation put $\tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right) = \phi$

$$\therefore \quad c(t) = \lim_{\zeta \to 1}\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n\sqrt{1-\zeta^2}\,t + \phi)\right]$$

or
$$c(t) = \lim_{\zeta \to 1}\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} [\sin(\omega_n\sqrt{1-\zeta^2}\,t)\cos\phi + \cos(\omega_n\sqrt{1-\zeta^2}\,t)\sin\phi]\right\}$$

Since, $\tan\phi = \frac{\sqrt{1-\zeta^2}}{\zeta}$, $\cos\phi = \zeta$ and $\sin\phi = \sqrt{1-\zeta^2}$ hence, substituting for $\cos\phi$ and $\sin\phi$ in the above equation

$$c(t) = \lim_{\zeta \to 1}\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} [\sin(\omega_n\sqrt{1-\zeta^2}\,t)\,.\,\zeta + \cos(\omega_n\sqrt{1-\zeta^2}\,t)\,.\,\sqrt{1-\zeta^2}\right\} \quad \text{...(6.26)}$$

Now $\lim_{\zeta \to 1} \sin(\omega_n\sqrt{1-\zeta^2}\,t)\,.\,\zeta \to \omega_n\sqrt{1-\zeta^2}\,t$ ...(6.27)

and $\lim_{\zeta \to 1} \cos(\omega_n\sqrt{1-\zeta^2}\,t) \to 1$ ...(6.28)

Substituting Eqs. (6.27) and (6.28) in Eq. (6.26),

$$c(t) = \lim_{\zeta \to 1}\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} (\omega_n\sqrt{1-\zeta^2}\,t + 1\,.\,\sqrt{1-\zeta^2})\right]$$

$$= \lim_{\zeta \to 1}\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sqrt{1-\zeta^2}\,(\omega_n t + 1)\right]$$

or
$$c(t) = [1 - e^{-\omega_n t}(1 + \omega_n t)] \quad \text{...(6.29)}$$

The time response in relation to Eq. (6.29) is plotted in Fig. 6.4.4. The response is called critically damped.

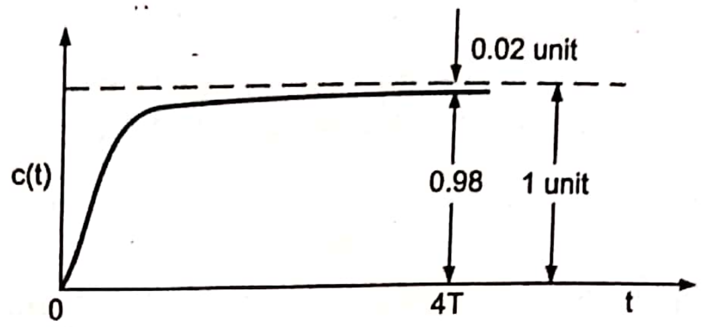

Fig. 6.4.4. Time response of a second order control system ($\zeta = 1$), critically damped case) subjected to unit step function.

(*d*) An expression for the time response of a second order control system having $\zeta > 1$ when subjected to unit step input function is derived hereunder:

The output for the system is given by

$$C(s) = R(s)\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad \text{...(6.30)}$$

As the input is a unit step function $r(t) = 1$ and $R(s) = 1/s$, therefore, substituting in Eq. (6.30)

$$C(s) = \frac{1}{s}\cdot\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad \text{...(6.31)}$$

Eq. (6.31) can also be written as

$$C(s) = \frac{1}{s}\cdot\frac{\omega_n^2}{(s + \zeta\omega_n)^2 - \omega_n^2(\zeta^2 - 1)}$$

or

$$C(s) = \frac{1}{s}\cdot\frac{\omega_n^2}{[s + (\zeta + \sqrt{\zeta^2 - 1})\,\omega_n][s + (\zeta - \sqrt{\zeta^2 - 1})\,\omega_n]} \qquad \text{...(6.32)}$$

Expanding R.H.S. of Eq. (6.32) into partial fractions,

$$C(s) = \frac{1}{s} - \frac{1}{2\sqrt{\zeta^2 - 1}\,(\zeta - \sqrt{\zeta^2 - 1})\,[s + (\zeta - \sqrt{\zeta^2 - 1})\omega_n]}$$

$$+ \frac{1}{2\sqrt{\zeta^2 - 1}\,(\zeta + \sqrt{\zeta^2 - 1})\,[s + (\zeta + \sqrt{\zeta^2 - 1})\omega_n]} \qquad \text{...(6.33)}$$

Taking inverse Laplace transform on both sides of equation (6.33)

$$c(t) = 1 - \frac{e^{-(\zeta - \sqrt{\zeta^2 - 1})\omega_n t}}{2\sqrt{\zeta^2 - 1}\,(\zeta - \sqrt{\zeta^2 - 1})} + \frac{e^{-(\zeta + \sqrt{\zeta^2 - 1})\omega_n t}}{2\sqrt{\zeta^2 - 1}\,(\zeta + \sqrt{\zeta^2 - 1})} \qquad \text{...(6.34)}$$

The time response in relation to Eq. (6.34) is plotted in Fig. 6.4.5. The response is called overdamped.

response of the system. When $\zeta = 1$ the actual damping equals ω_n. This actual damping when $\zeta = 1$ is known as critical damping. At the value of critical damping the oscillations in the time response just disappear.

The ratio of the actual damping to critical damping is known as the damping ratio as given by the following relation:

$$\frac{\text{Actual damping}}{\text{Critical damping}} = \frac{\zeta\omega_n}{\omega_n} = \zeta$$

6.4.3. Characteristic Equation

As mentioned earlier a general expression for the transfer function of a second order control system is given by

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad \text{...(6.35 a)}$$

The denominator of the above expression is equated to zero and following equation is formed

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad \text{...(6.36)}$$

The above equation is quadratic in s and two roots are

$$s_1, s_2 = -\zeta\omega_n \pm j\omega_n \sqrt{1-\zeta^2}$$

The roots s_1, s_2 are also the poles of transfer function expressed by relation (6.35 a). A study of roots s_1, s_2 gives a prediction about the nature of time response. The real part of the roots represents the damping and the imaginary part represents damped frequency of oscillations. Therefore, equation

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

is called the characteristic equation of a second order control system.

The location of roots of the characteristic equation for various values of ζ (keeping ω_n fixed) and the corresponding time response for a second order control system is shown in Fig. 6.4.7.

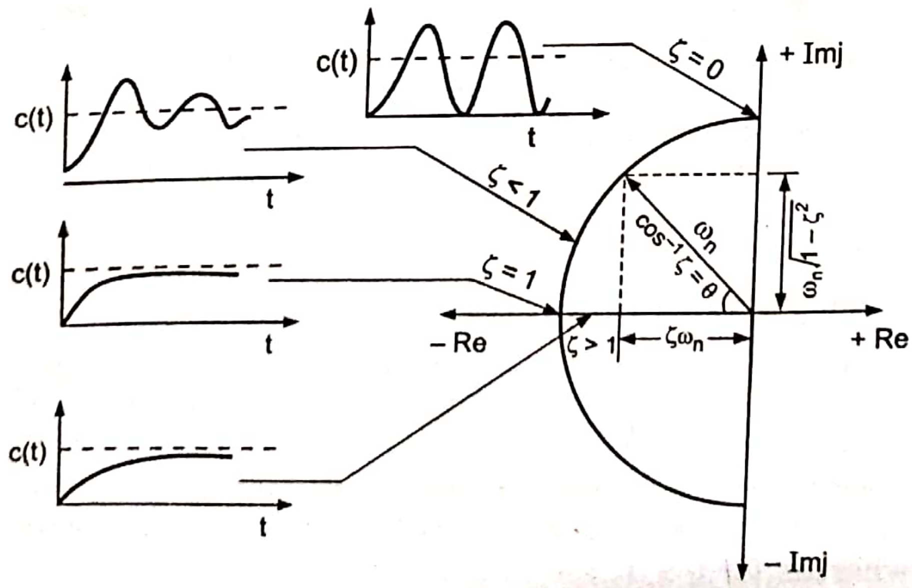

Fig. 6.4.7. Location of roots of the characteristic equation and corresponding time response.

The coefficients K_1, K_2 and K_3 can be determined as

$$K_1 = 1,\ K_2 = -2 \text{ and } K_3 = 1$$

$$\therefore \qquad C(s) = \frac{1}{s} - \frac{2}{(s+1)} + \frac{1}{(s+2)}$$

Taking inverse Laplace transform on both sides $c(t) = (1 - 2e^{-t} + e^{-2t})$. **Ans.**

Example 6.10.4. *Fig. 6.10.2 represents a control system. Obtain an expression relating the output and time. The input being unit step.*

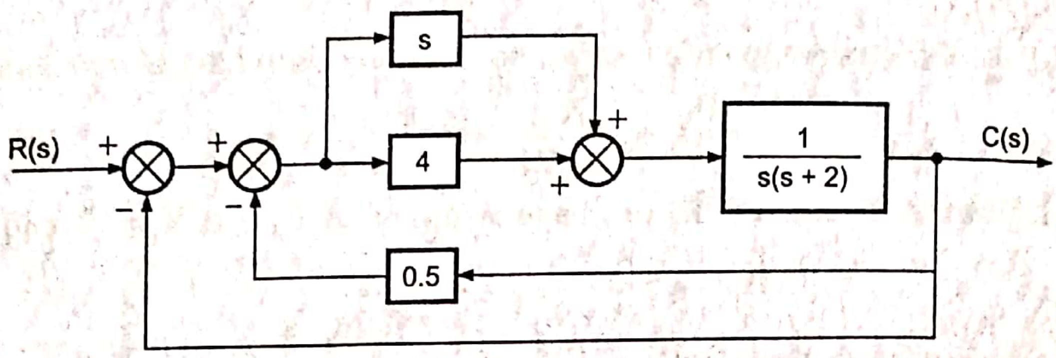

Fig. 6.10.2. Control system for example 6.10.2 (*a*).

Solution. The block diagram of Fig. 6.10.2 is reduced in steps below :

From the block diagram of Fig. 6.10.2 (*b*) the transfer function of the system is determined below :

$$\frac{C(s)}{R(s)} = \frac{\dfrac{(s+4)}{(s^2+2.5s+2)}}{1 + \dfrac{(s+4)}{(s^2+2.5s+2)}\cdot 1} \quad \text{or} \quad \frac{C(s)}{R(s)} = \frac{(s+4)}{(s^2+3.5s+6)}$$

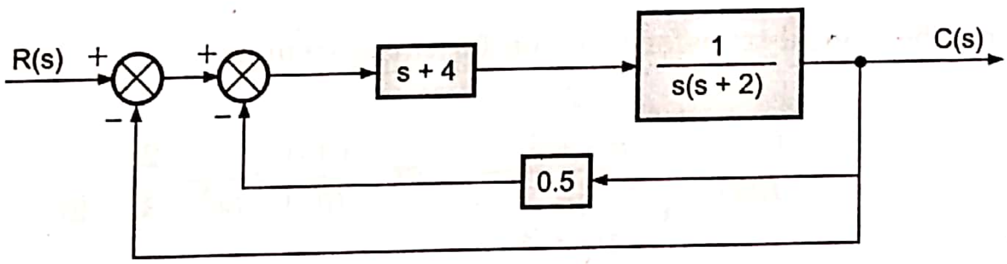

Fig. 6.10.2 (*a*) Block diagram reduction of Fig. 6.10.2.

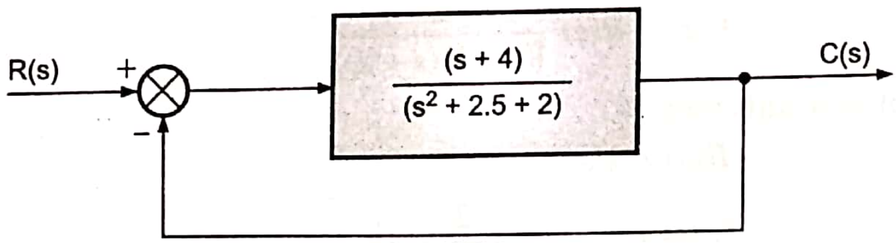

Fig. 6.10.2 (*b*) Block diagram reduction of Fig. 6.10.2 (*a*).

$$\therefore \qquad C(s) = R(s)\frac{(s+4)}{(s^2+3.5s+6)}$$

As $\qquad r(t) = 1,\ R(s) = \dfrac{1}{s}$

$$\therefore \quad C(s) = \frac{1}{s} \cdot \frac{(s+4)}{(s^2 + 3.5s + 6)}$$

The R.H.S. of the above equation can be expanded into partial fraction

$$\therefore \quad \frac{1}{s} \cdot \frac{(s+4)}{(s^2+3.5s+6)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{(s^2+3.5s+6)}$$

$$K_1 = 2/3, K_2 = -2/3 \text{ and } K_3 = -4/3$$

$$\therefore \quad C(s) = \frac{2}{3} \cdot \frac{1}{s} - \frac{2}{3} \cdot \frac{s}{(s^2+3.5s+6)} - \frac{4}{3} \cdot \frac{1}{(s^2+3.5s+6)}$$

$$= \frac{2}{3} \cdot \frac{1}{s} - \frac{2}{3} \cdot \frac{s}{[(s^2+3.5s+(1.75)^2+6-(1.75)^2]} - \frac{4}{3} \cdot \frac{1}{[(s^2+3.5s+(1.75)^2+6-(1.75)^2]}$$

$$= \frac{2}{3} \cdot \frac{1}{s} - \frac{2}{3} \cdot \frac{s}{[(s+1.75)^2+(1.71)^2]} - \frac{4}{3} \cdot \frac{1}{[(s+1.75)^3+(1.71)^2]}$$

$$= \frac{2}{3} \cdot \frac{1}{s} - \frac{2}{3} \cdot \frac{[(s+1.75)-1.75]}{[(s+1.75)^2+(1.71)^2]} - \frac{4}{3} \cdot \frac{1}{[(s+1.75)^2+(1.71)^2]}$$

$$= \frac{2}{3} \cdot \frac{1}{s} - \frac{2}{3} \cdot \frac{(s+1.75)}{[(s+1.75)^2+(1.71)^2]} + \frac{2}{3} \cdot \frac{1.75}{[(s+1.75)^2+1.71^2]}$$

$$- \frac{4}{3} \cdot \frac{1}{[(s+1.75)^2+(1.71)^2]}$$

$$= \frac{2}{3} \cdot \frac{1}{s} - \frac{2}{3} \cdot \frac{(s+1.75)}{[(s+1.75)^2+1.71^2]} - \frac{0.5}{3} \cdot \frac{1}{[(s+1.75)^2+(1.71)^2]}$$

$$= \frac{2}{3} \cdot \frac{1}{s} - \frac{2}{3} \cdot \frac{(s+1.75)}{[(s+1.75)^2+(1.71)^2]} - \frac{0.5}{3 \times 1.71} \cdot \frac{1.71}{[(s+1.75)^2+(1.71)^2]}$$

Taking inverse Laplace transform

$$c(t) = \frac{2}{3} - \frac{2}{3} e^{-1.75t} . \cos(1.71t) - \frac{0.29}{3} . e^{-1.75t} . \sin(1.71t). \qquad \textbf{Ans.}$$

Example 6.10.5. *The block diagram of a unity feedback control system is shown in Fig. 6.10.3.*

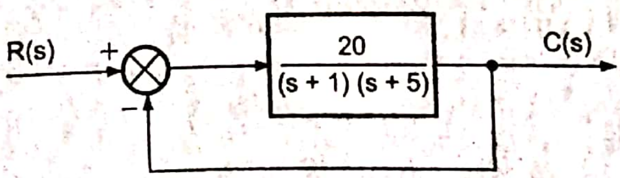

Fig. 6.10.3. Block diagram for example 6.10.5.

Determine the characteristic equation of the system, ω_n, ζ, ω_d, t_p, M_p, the time at which the first undershoot occurs, the time period of oscillations and the number of cycles completed before reaching the steady state.

Solution. The overall transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{\dfrac{20}{[(s+1)(s+5)]}}{1+\dfrac{20}{[(s+1)(s+5)]}\cdot 1} = \frac{20}{(s^2+6s+25)} = \frac{20}{25}\cdot\frac{25}{(s^2+6s+25)}$$

The characteristic equation is given by

$$(s^2 + 6s + 25) = 0. \quad \textbf{Ans.}$$

From the characteristic equation following quantities are calculated :

$$\omega_n = \sqrt{25} = 5 \text{ rad/sec.} \quad \textbf{Ans.}$$

$$2\,\zeta\,\omega_n = 6$$

$$\therefore \qquad \zeta = \frac{6}{2\omega_n} = \frac{6}{2\times 5} = 0.6. \quad \textbf{Ans.}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 5\sqrt{1-(0.6)^2} = 4 \text{ rad/sec.} \quad \textbf{Ans.}$$

The time at which the maximum overshoot occurs is

$$t_p = \frac{\pi}{\omega_n\left(\sqrt{1-\zeta^2}\right)} = \frac{\pi}{5\sqrt{1-0.6^2}} = 0.78 \text{ sec.} \quad \textbf{Ans.}$$

The maximum overshoot is given by

$$M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}\times 100 = e^{-\frac{0.6\pi}{\sqrt{1-0.6^2}}}\times 100 = 9.4\%. \qquad \textbf{Ans.}$$

The time at which the first undershoot occurs is

$$t = \frac{2\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{2\pi}{5\sqrt{1-(0.6)^2}} = 1.56 \text{ sec.} \quad \textbf{Ans.}$$

$$\text{The period of oscillation} = \frac{2\pi}{\omega_d} = \frac{2\pi}{4} = 1.56 \text{ sec.} \quad \textbf{Ans.}$$

As the steady state is reached in time $4/\zeta\omega_n$, the number of oscillations completed before reaching steady state is

$$\frac{\omega_d}{2\pi}\times\frac{4}{\zeta\omega_n} = \frac{4}{2\pi}\times\frac{4}{0.6\times 5} = 0.84 \text{ cycle.} \quad \textbf{Ans.}$$

Example 6.10.6. *A load is controlled by an electric motor. The position of the input shaft is adjusted to θ_r. The motor torque constant is 360 Nm per radian of error between the input and output position. The moment of inertia and the coefficient of viscous friction are 10 kg-m^2 and 60 Nm/(rad/sec) respectively. The system being initially at rest. Determine an expression relating time and angular position of the output following a sudden shift of 50° in the input shaft.*

Solution. The block diagram for the system is shown in Fig. 6.10.4.

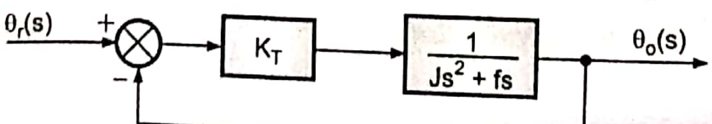

Fig. 6.10.4. Control system for example 6.10.6.

Example 6.10.15. *The open-loop transfer function of a unity feedback control system is given by:*

$$G(s) = \frac{K(s+2)}{s^3 + \beta s^2 + 4s + 1}$$

Determine the value of K and β such that the closed-loop unit step response has $\omega_n = 3 \text{ rad/sec}$ *and* $\zeta = 0.2$.

Solution. The characteristics equation of the system is

$$1 + G(s) = 0$$

$$\therefore \quad 1 + \frac{K(s+2)}{s^3 + \beta s^2 + 4s + 1} = 0$$

or $$s^3 + \beta s^2 + 4s + 1 + K(s+2) = 0$$

or $$s^3 + \beta s^2 + (K+4)s + (2K+1) = 0$$

Let $$(s+a)(s^2 + bs + c) = 0 \quad ...(1)$$

$\therefore$ $$s^3 + (a+b)s^2 + (ab+c)s + ac = 0 \quad ...(2)$$

It is known that the characteristic equation of a second order system is given by ...(3)

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad ...(4)$$

Comparing coefficients of second term of Eq. (2) and that of (4), the values of b and c are obtained below :

$$c = \omega_n^2 = (3)^2 = 9$$

$$b = 2\zeta\omega_n = 2 \times 0.2 \times 3 = 1.2$$

On comparing coefficients of Eq. (1) and (3)

$$(a+b) = \beta \quad ...(5)$$

$$(ab+c) = K+4 \quad ...(6)$$

$$ac = 2K+1 \quad ...(7)$$

From equation (7) : $a = \dfrac{2K+1}{c} \quad \because \quad c = 9$

$\therefore$ $$a = (2K+1)/9$$

Since, $$b = 1.2, \text{ and } c = 9$$

$$(ab+c) = K+4$$

$\therefore$ $$(a \times 1.2 + 9) = K+4$$

$\because$ $$a = (2K+1)/9$$

$\therefore$ $$[(2K+1)/9] \times 1.2 + 9 = K+4 \quad \text{or} \quad 6.6K = 45$$

$$K = \frac{45}{6.6} = 6.81. \quad \textbf{Ans.}$$

$\therefore$ $$a = \frac{2K+1}{9} = \frac{2 \times 6.81 + 1}{9} = 1.51$$

From equation (5) $\quad \beta = (a+b) = (1.51 + 1.2) = 2.71.$ **Ans.**

Example 6.10.16. *Determine the damping ratio and undamped natural frequency of oscillatory roots and $\%M_p$ for a unit step input given that :*

$$\frac{C(s)}{E(s)} = \frac{1}{s(1+0.5s)(1+0.2s)}$$

and the system is unity feedback type.

Solution. $$\frac{C(s)}{E(s)} = \frac{1}{0.5 \times 0.2 s(s+2)(s+5)} = \frac{10}{s(s+2)(s+5)} = \frac{10}{s^3 + 7s^2 + 10s}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s) \times 1}$$

$\because$ $$G(s) = \frac{10}{s^3 + 7s^2 + 10s}$$

$$\frac{C(s)}{R(s)} = \frac{\frac{10}{s^3 + 7s^2 + 10s}}{1 + \frac{10}{s^3 + 7s^2 + 10s} \times 1} \quad \text{or} \quad \frac{C(s)}{R(s)} = \frac{10}{s^3 + 7s^2 + 10s + 10}$$

The characteristic equation of the system is given by

$$s^3 + 7s^2 + 10s + 10 = 0$$

To a fair approximation one of the three roots of the above characteristic equation can be determined as $s = -5.5$, thus the quadratic term of the above equation can be determined as

$$\frac{s^3 + 7s^2 + 10s + 10}{s + 5.5} = s^2 + 1.5s + 1.8$$

$$\therefore \qquad \frac{C(s)}{R(s)} = \frac{10}{(s + 5.5)(s^2 + 1.5s + 1.8)}$$

The characteristic equation of the system can be rewritten as

$$(s + 5.5)\,(s^2 + 1.5s + 1.8) = 0$$

For oscillatory roots :

$$\omega_n^2 = 1.8$$

or $$\omega_n = \sqrt{1.8} = 1.34 \text{ rad/sec.} \quad \textbf{Ans.}$$

or $$2\zeta\omega_n = 1.5$$

$$\therefore \qquad \zeta = \frac{1.5}{2\omega_n} = \frac{1.5}{2 \times 1.34} = 0.56. \quad \textbf{Ans.}$$

The time constant of quadratic term is $1/\zeta\omega_n = \frac{1}{0.56 \times 1.34} = 1.33$ sec

The time constant due to first order term is $1/5.5 = 0.18$ sec, its effect is neglected as compared to quadratic term time constant 1.33 sec

$$\%M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-0.56\pi}{\sqrt{1-0.56^2}}} \times 100$$

$$= e^{-\frac{1.76}{0.83}} \times 100 = e^{-2.12} \times 100 = 11.9\%. \quad \textbf{Ans.}$$

Example 6.10.17. *The open-loop transfer function of a control system is given below:*

$$G(s)\,H(s) = \frac{2(s^2 + 3s + 20)}{s(s + 2)(s^2 + 4s + 10)}$$

Determine the static error coefficients and steady state error for the input given as:

(a) 5 *(b)* $4t$ *(c)* $4t^2/2$.